# CODECHECK certificate 2026-002

https://codecheck.org.uk/register/certs/2026-002/

## Table 1: CODECHECK summary

| Item | Value |
|---|---|
| Title | *Preserving privacy of spatial distances using randomized geometric surface calculations* |
| Author(s) | Jonas Klingwort, Sarah Redlich |
| Reference | https://doi.org/10.5311/JOSIS.2025.31.375 |
| Repository | https://github.com/jkwort/ppdc |
| Codechecker(s) | Linus Dexter Hackel |
| Date of check | 2026-06-29 |
| Summary | TODO add summary |

## Table 2: Summary of output files generated

| File | Comment | Size (b) |
|---|---|---|
| `figure_1_step_1.pdf` | Figure 1 - Step 1 - Generate base map with Germany and bounding box | 263198 |
| `figure_1_step_2.pdf` | Figure 1 - Step 2 - Calculate true distances | 263388 |
| `figure_1_step_3.pdf` | Figure 1 - Step 3 - Draw five random coordinates and obtain triangles | 263972 |
| `figure_5.pdf` | Figure 5 - Relative root mean squared error split by country (Germany in green, the Nether-lands in orange). | 7468 |
| `figure_6_1.pdf` | Figure 6 - Convergence plots based on 1,000 Monte Carlo simulations. This shows the German dataset. The X-axis shows the number of samples, and the y-axis shows the MC mean. This is the German dataset and 300 random points. The green line shows the point estimate, and the gray lines show the lower and upper confidence interval limits. | 16956 |
| `figure_6_2.pdf` | Figure 6 - Convergence plots based on 1,000 Monte Carlo simulations. This shows the Dutch dataset. The X-axis shows the number of samples, and the y-axis shows the MC mean. This is the Dutch dataset and one random point. The orange line shows the point estimate, and the gray lines show the lower and upper confidence interval limits. | 16449 |

| File | Comment | Size (b) |
|---|---|---|
| `figure_8.pdf` | Figure 8 - Correlation coefficient between Haversine distance and different versions of theproxy. The x-axis shows the number of randomly sampled points used to obtain the proxy and the y-axis shows the correlation coefficient. The color and shapes indicate the version of the proxy information. | 8846 |
| `table_1.csv` | Table 1 - Calculate surface areas of the triangles | 226 |
| `table_2.csv` | Table 2 - Distribution of geographical distances (Haversine distance in km) for Germany and the Netherlands. | 176 |

## Summary

TODO add summary

## CODECHECKER notes

For a majority of the scripts included in the `code` directory of the code repository, the data first had to be created with the script: `03_section_6_prep_evaluation.R` which is provided in the `code` directory as well. The problem is that the authors used a large CPU HPC cluster to calculate these datasets, which I as a codechecker don't have access to. But since the datasets are created in the loop beginning in line 79, this loop can also be stopped earlier to produce a portion of the needed base datasets. After consulting with Daniel Nüst, we decided that the portions of the base datasets would still be sufficient to check the reproducibility of the code.

When reproducing Table 2, I noticed that the values are all equal to the paper, but the Header of the Table isn't matching. This could either be down to me using the reduced base datasets for reproduction, or the authors later renaming the Column Headers in their paper. My column headers are: `Country, Min., 1^st Quantile, Median, Mean, 3^rd Quantile, Maximum` , while the papers column headers are: `Country, Min., 25^th Quartile, 50^th Quartile, Mean, 75^th Quartile, Max.` .

## Recommendations to the authors

*TODO*

## Manifest files

### CSV files

`table_1.csv`

Author comment: *Table 1 - Calculate surface areas of the triangles*

| R_n | base_side_of_triangle | height_of_the_base_side | surface_area |
|---|---|---|---|
| 1 | 1.36115 | 1.75236 | 1.19261 |
| 2 | 1.36115 | 5.52275 | 3.75864 |
| 3 | 1.36115 | 2.72697 | 1.85591 |
| 4 | 1.36115 | 3.65482 | 2.48738 |
| 5 | 1.36115 | 2.67144 | 1.81811 |

`table_2.csv`

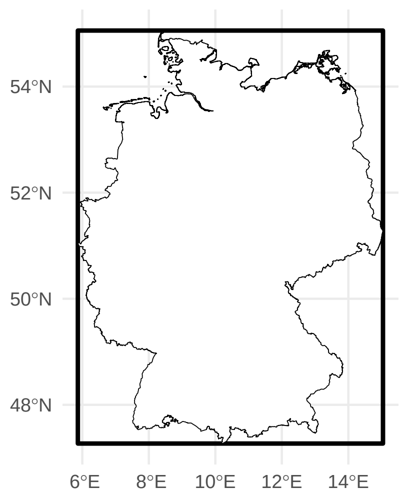
Author comment: *Table 2 - Distribution of geographical distances (Haversine distance in km) for Germany and the Netherlands.*

| Country | Min. | 1^st Quantile | Median | Mean | 3^rd Quantile | Maximum |
|---|---|---|---|---|---|---|
| Germany | 0.11 | 199.62 | 315.03 | 320.39 | 431.35 | 875.67 |
| Netherlands | 0.01 | 53.9 | 88.03 | 95.35 | 131.24 | 323.88 |

### Figures

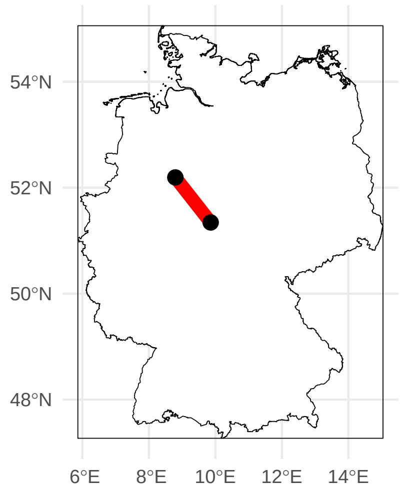
`figure_1_step_1.pdf`

Author comment: *Figure 1 - Step 1 - Generate base map with Germany and bounding box*



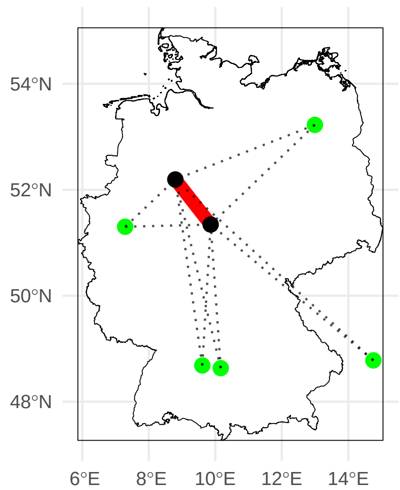

`figure_1_step_2.pdf`

Author comment: *Figure 1 - Step 2 - Calculate true distances*



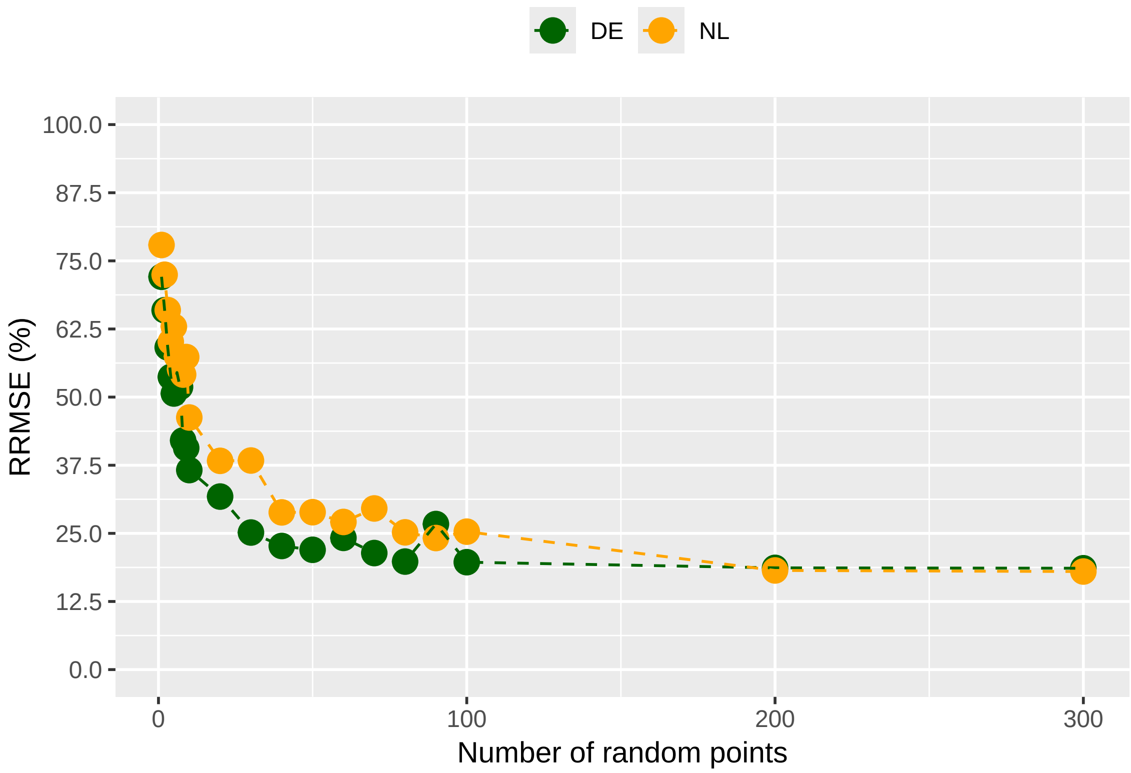

`figure_1_step_3.pdf`

Author comment: *Figure 1 - Step 3 - Draw five random coordinates and obtain triangles*



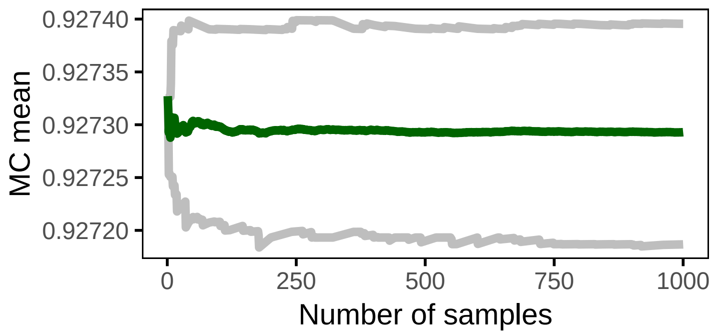

`figure_5.pdf`

Author comment: *Figure 5 - Relative root mean squared error split by country (Germany in green, the Nether-lands in orange).*



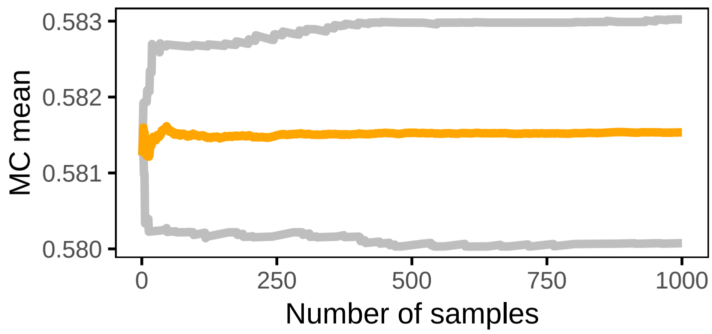

`figure_6_1.pdf`

Author comment: *Figure 6 - Convergence plots based on 1,000 Monte Carlo simulations. This shows the German dataset. The X-axis shows the number of samples, and the y-axis shows the MC mean. This is the German dataset and 300 random points. The green line shows the point estimate, and the gray lines show the lower and upper confidence interval limits.*



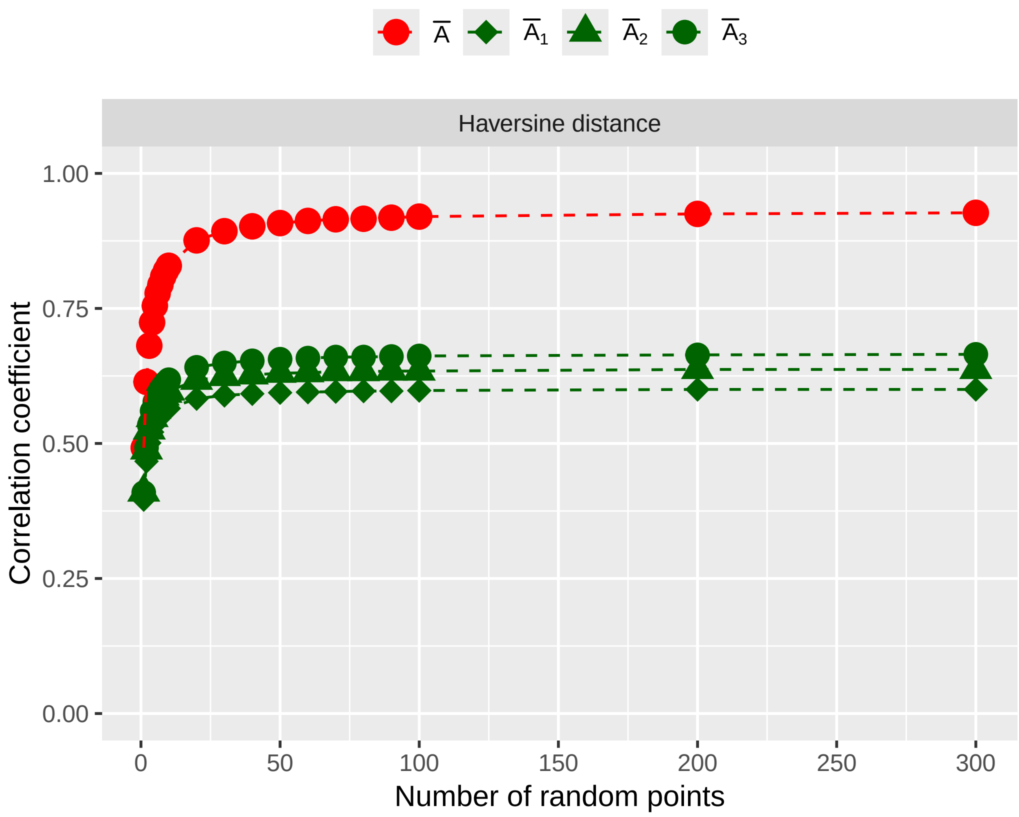

`figure_6_2.pdf`

Author comment: *Figure 6 - Convergence plots based on 1,000 Monte Carlo simulations. This shows the Dutch dataset. The X-axis shows the number of samples, and the y-axis shows the MC mean. This is the Dutch dataset and one random point. The orange line shows the point estimate, and the gray lines show the lower and upper confidence interval limits.*




`figure_8.pdf`

Author comment: *Figure 8 - Correlation coefficient between Haversine distance and different versions of theproxy. The x-axis shows the number of randomly sampled points used to obtain the proxy and the y-axis shows the correlation coefficient. The color and shapes indicate the version of the proxy information.*



## Acknowledgements

*TODO: acknowledge the authors who helped in reproducing the results and figures of the paper.*

CODECHECK is financially supported by the Mozilla foundation.

## Citing this document

Linus Dexter Hackel (2026). CODECHECK Certificate 2026-002. Zenodo. https://codecheck.org.uk/register/certs/2026-002/

## About CODECHECK

This certificate confirms that the codechecker could independently reproduce the results of a computational analysis given the data and code from a third party. A CODECHECK does not check whether the original computation analysis is correct. However, as all materials required for the reproduction are freely availableby following the links in this document, the reader can then study for themselves the code and data.

## About this document

This document was created using codecheck-py (a Python-base template for creating CODECHECK certificates). The CODECHECK details are filled into a jupyter notebook which is then converted into Markdown via nbconvert. Afterwards it gets converted into Typst using cmarker and then into PDF using Typst. `sh notebook_to_pdf.sh` will regenerate the report file.

```
import session_info2 as si
si.session_info(os=True, cpu=True, gpu=True, dependencies=True)
```

```
traitlets 5.14.3
PyYAML  6.0.3
certifi 2026.1.4 (2026.01.04)
requests  2.32.5
six 1.17.0
tornado 6.5.3
jupyter_client  8.7.0
pure_eval 0.2.3
prompt_toolkit  3.0.52
pytz  2025.2
setuptools  80.9.0
session-info2 0.3
decorator 5.2.1
debugpy 1.8.17
psutil  7.1.3
packaging 25.0
idna  3.11
comm  0.2.3
charset-normalizer  3.4.4
python-dateutil 2.9.0.post0
tabulate  0.9.0
PySocks 1.7.1
urllib3 2.6.1
Pygments  2.19.2
numpy 2.3.5
stack_data  0.6.3
pandas  3.0.0
wcwidth 0.2.14
platformdirs  4.5.1
asttokens 3.0.1
pyzmq 27.1.0
ipykernel 7.1.0
Brotli  1.2.0
```

```
parso 0.8.5
jupyter_core  5.9.1
jedi  0.19.2
colorama  0.4.6
ipython 9.8.0
executing 2.2.1
----  ----
Python  3.14.2 | packaged by conda-forge | (main, Dec  6 2025, 11:21:58) [GCC 14.3.0]
OS  Linux-6.18.33.2-microsoft-standard-WSL2-x86_64-with-glibc2.35
CPU 16 logical CPU cores, x86_64
GPU No GPU found
Updated 2026-06-29 15:22
```

## License